Renewable Energy and Carbon Emissions

Global Panel Analysis, 1991-2022 | Individual Stata Project

Project objective

This project examines how renewable energy consumption is associated with per-capita CO_2 emissions after accounting for economic development and time-invariant differences across economies.

Workflow

Collect four WDI indicators -> reshape and validate keys -> merge by country code and year -> estimate panel models -> interpret and document findings

Data and methods

Component	Portfolio evidence
Source	World Bank World Development Indicators
Original analytical sample	6,720 complete economy-year observations; 1991-2022
Variables	CO_2 per capita, renewable-energy share, services share, GDP per capita (PPP)
Main method	Country fixed effects with country-clustered robust standard errors
Sensitivity checks	Panel lags, winsorization, subgroup comparisons, quadratic specification

Selected results

Specification	Renewable coefficient	Interpretation
Baseline, no controls	0.0396**	Positive conditional association
Full fixed-effects model	0.0129	Small and statistically insignificant
Low-penetration subsample	0.0158	Statistically insignificant
High-penetration subsample	-0.2371***	Negative exploratory subgroup estimate

Interpretation

After controlling for GDP per capita and services share, renewable energy was not significantly associated with per-capita CO_2 emissions in the full sample. GDP per capita remained the strongest measured correlate. The contrast across penetration groups suggests possible heterogeneity, but it is not sufficient to establish a causal threshold.

Quality-control revisions

For the portfolio version, panel datasets are merged using country code and year; panel lags use Stata's L. operators; and the calculated 2.83% turning point is not presented as a confirmed threshold because the quadratic terms were statistically insignificant.

Skills demonstrated

Stata | World Bank data | Wide-to-long reshaping | Panel integration | Data validation | Fixed effects | Robustness checks | Statistical interpretation | Research writing

Limitations

The analysis is observational, uses an unbalanced panel, and may include World Bank aggregate entities. It does not fully address reverse causality or omitted time-varying confounders. Subgroup results are exploratory rather than a formal panel-threshold estimate.